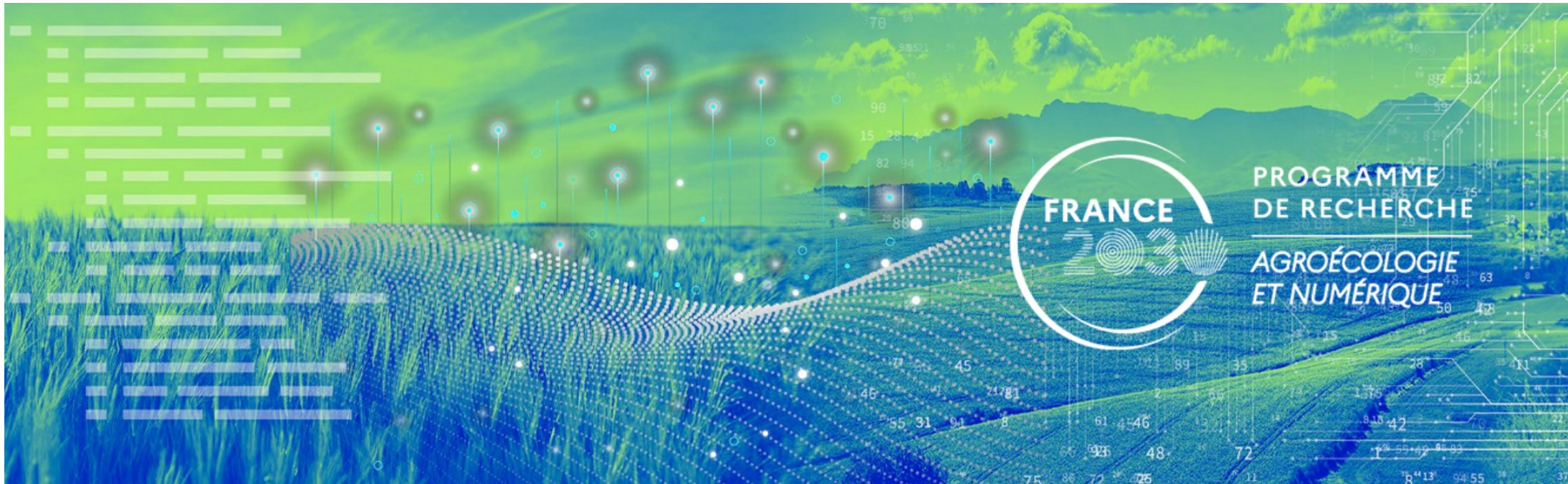




Building the LoRa IoT platform

Part 2: the customized IoT gateway

Prof. Congduc Pham
<http://www.univ-pau.fr/~cpham>
 Université de Pau, France

Congduc.Pham@univ-pau.fr



Wireless Sensors Made Simple
 for agroecology & sustainable agriculture

Powered by technologies developed in  Intel-IrriS



The customized IoT gateway

- ⦿ The customized IoT gateway is based on the generic WaziGate
- ⦿ WaziGate is an IoT LoRa Gateway developed by WAZIUP
- ⦿ WaziGate implements the edge-enabled IoT gateway approach
 - ⦿ customized applications can be directly hosted in the gateway
 - ⦿ the gateway can easily work without Internet connectivity
 - ⦿ data are available to end-users in an embedded database
 - ⦿ web-based visualization module provides graphical user interface
- ⦿ You can find all the WaziGate documentation on the [WaziGate documentation page](#).
- ⦿ The customized IoT gateway adds many more features
 - ⦿ Simple configuration for all supported sensor types
 - ⦿ Sensor dashboard based on Home Assistant
 - ⦿ Automatic backup and simple restore procedure for sensor data
 - ⦿ Display on OLED screen, ...



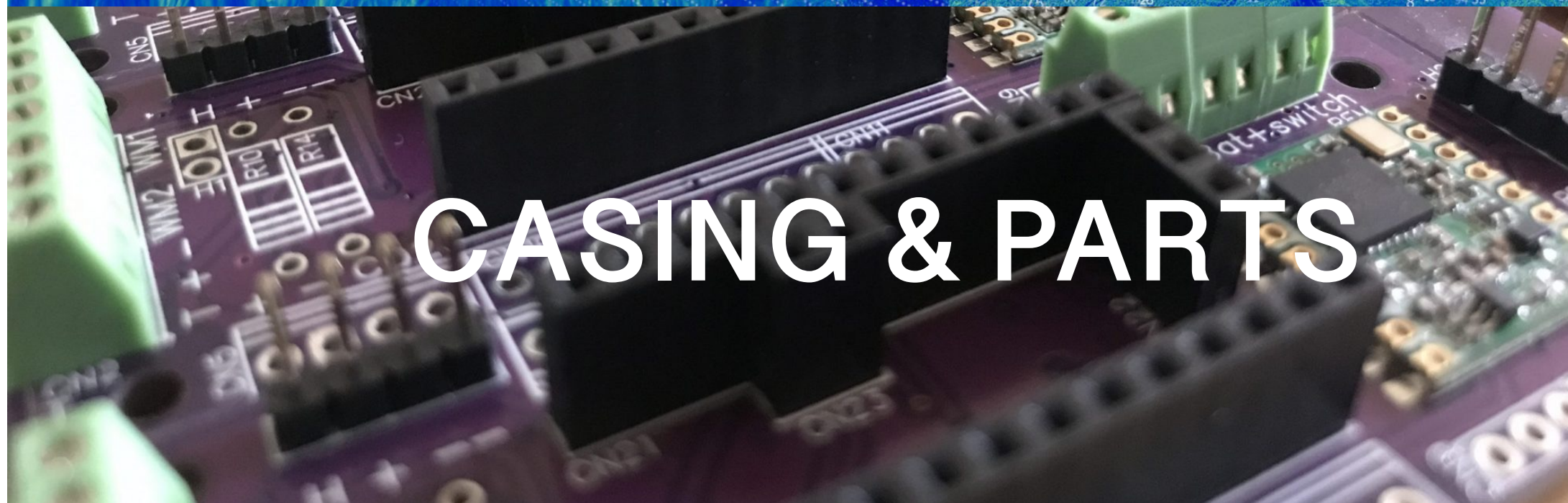
Install software for your gateway

⦿ PEPR AgriFutur gateway distribution (RPI3B/3B+/4B)

- ⦿ comes pre-configured
- ⦿ will work out-of-the box with the default sensor devices
- ⦿ Including the LoRaCAM-AI device
- ⦿ **Download the gateway SD card image from project website**
- ⦿ <http://intel-irris.eu/results>
- ⦿ Image uses EU868 frequency band
- ⦿ EU433 or other bands can be configured afterwards
- ⦿ Flash the SD card image on 8GB SD card class 10



Look at the video produced for INTEL-IRRIS for the main steps
Video at t=124s: <https://youtu.be/j-1Nk0tv0xM?t=124>





Raspberry PI enclosure

- ⦿ You can take a simple commercial plastic RPI case to easily drill holes for the antenna connector, or use a 3D printed enclosure

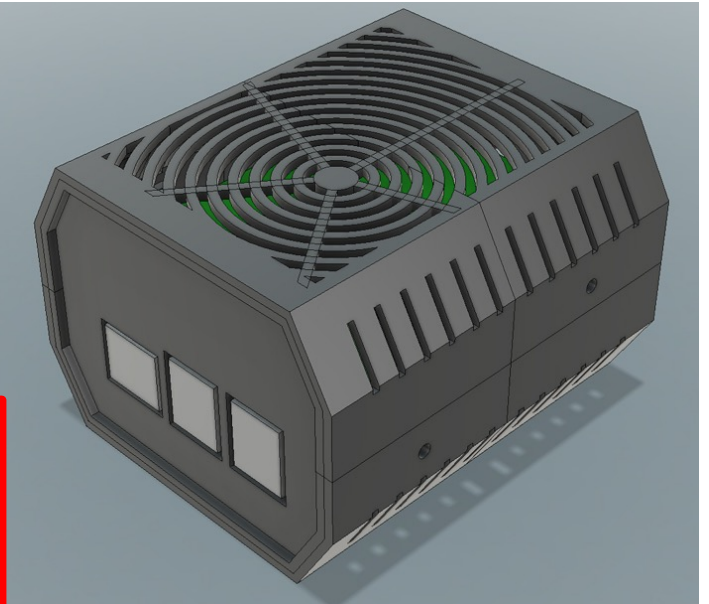
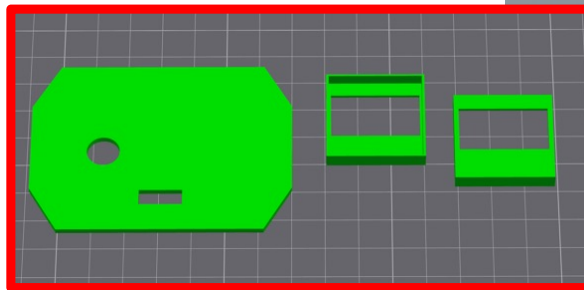
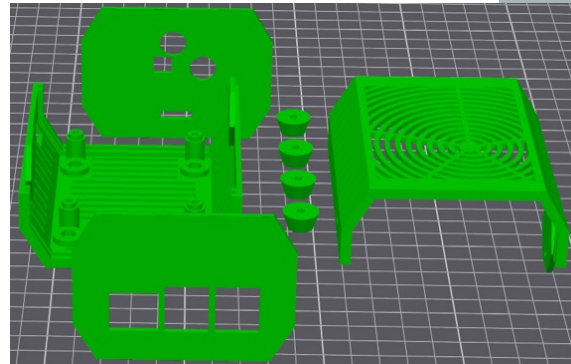


RPI3B/3B+



RPI4B

3D printed from <https://www.thingiverse.com/thing:3012012>



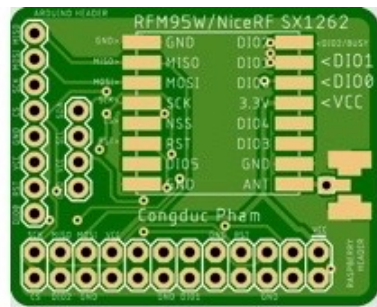
We provide modified & additional parts on GitHub



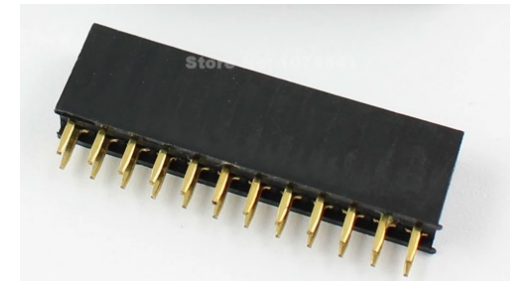
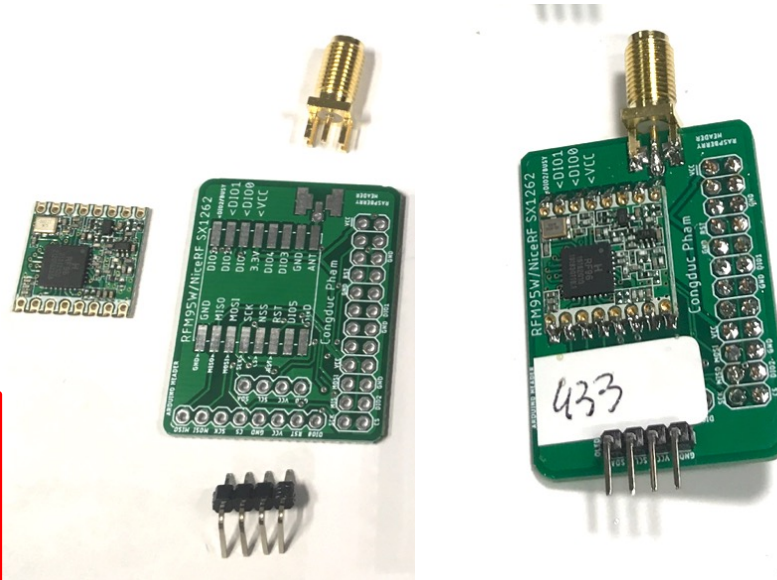
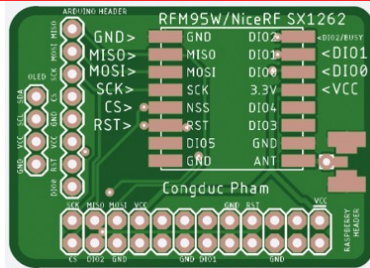
Single-channel: the LoRa radio hat

Information on how to manufacture and order a PCB:

<https://github.com/CongducPham/PRIMA-Intel-IrriS/blob/main/Tutorials/Intel-Irris-PCB-update-PCBA.pdf>



RFM95W/NiceRF SX1262
breakout



2-row 12-pin header
or
2 X 1-row 12-pin header

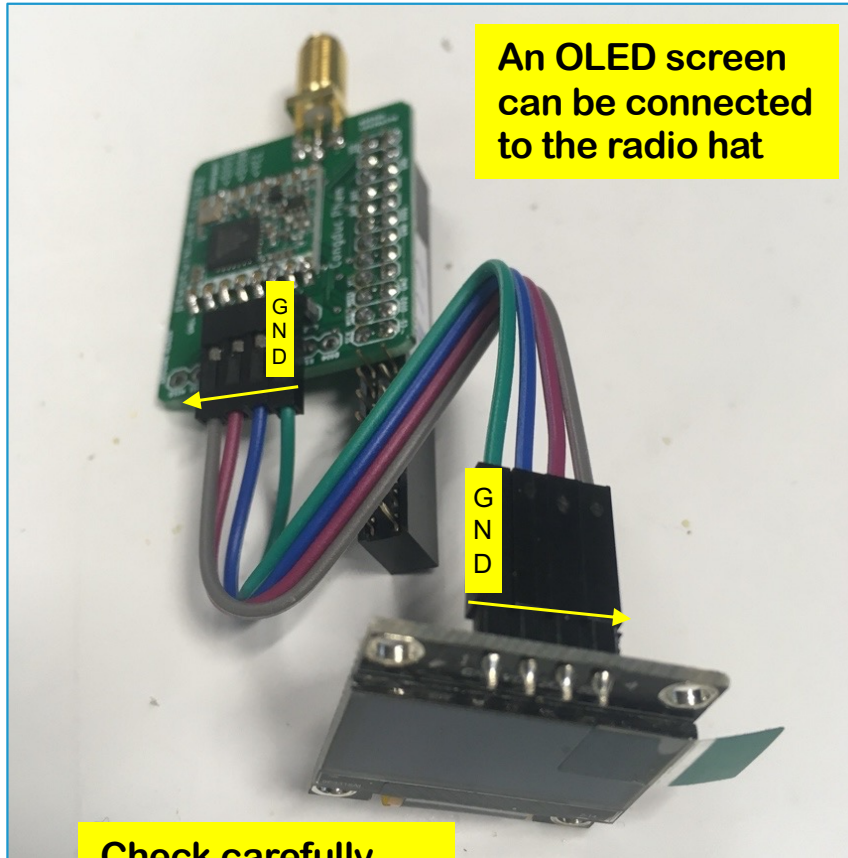


Get the zipped Gerber file from:

https://github.com/CongducPham/PEPR_AgriFutur/blob/main/PCBs/RFM95Breakout_2020-11-13-long.zip



Installing OLED



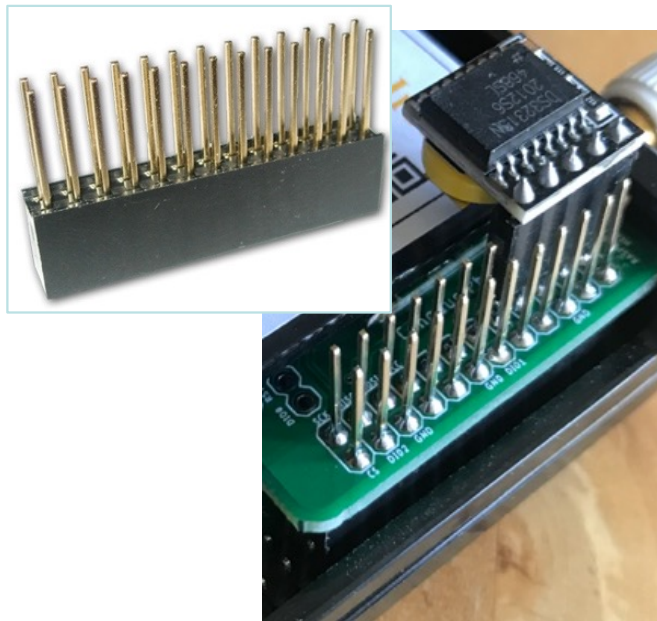
Check carefully
the pin order
which is reversed



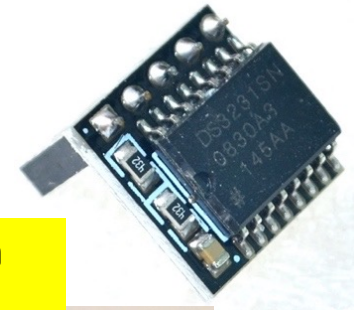
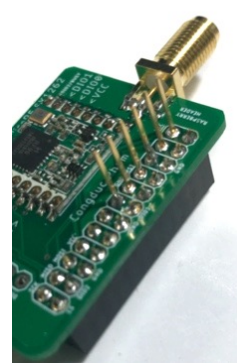
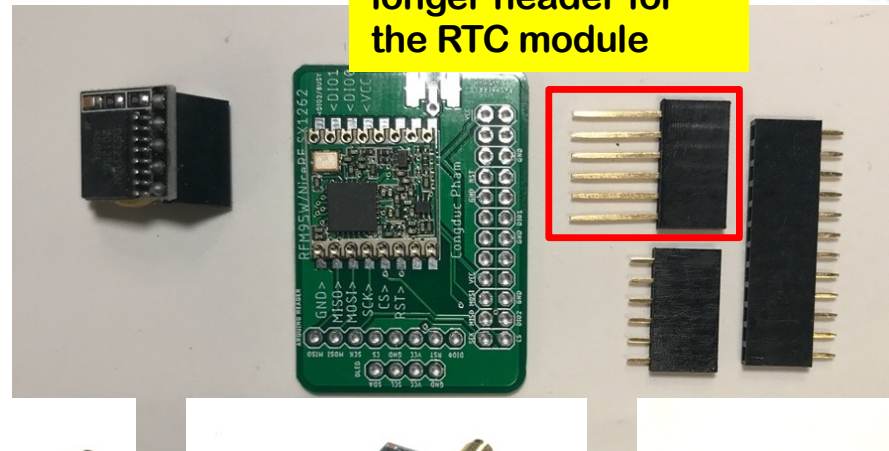


Connecting an RTC module

- Several possibilities depending on availability of components



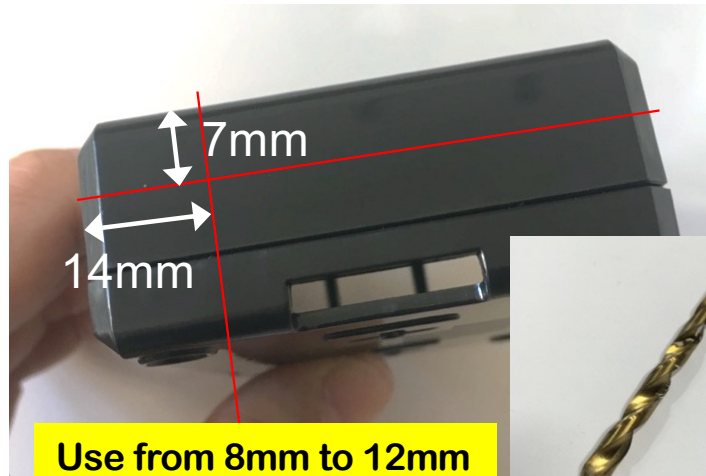
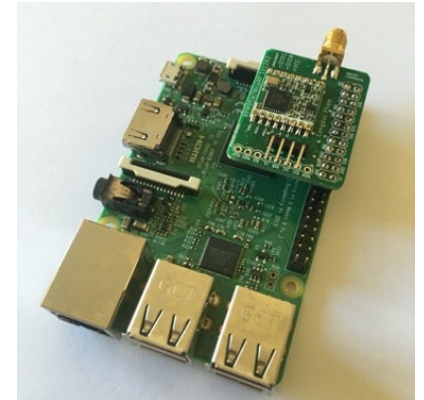
1/ Solder a header with longer pins on the radio hat. You can then cut the remaining pins if you want



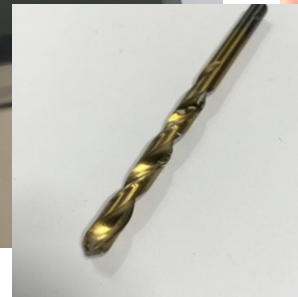


Preparing the enclosure

- ⦿ Recommended RPI model is RPI3B
 - ⦿ plug in the radio LoRa hat
 - ⦿ The OLED screen is optional but highly recommended
- ⦿ Drill a hole for the SMA connector on the RPI enclosure, at the SD card slot side
- ⦿ <https://fr.aliexpress.com/item/32718435597.html>



Use from 8mm to 12mm
drill bit for metal

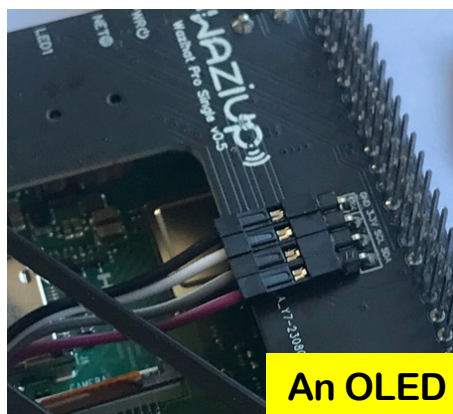




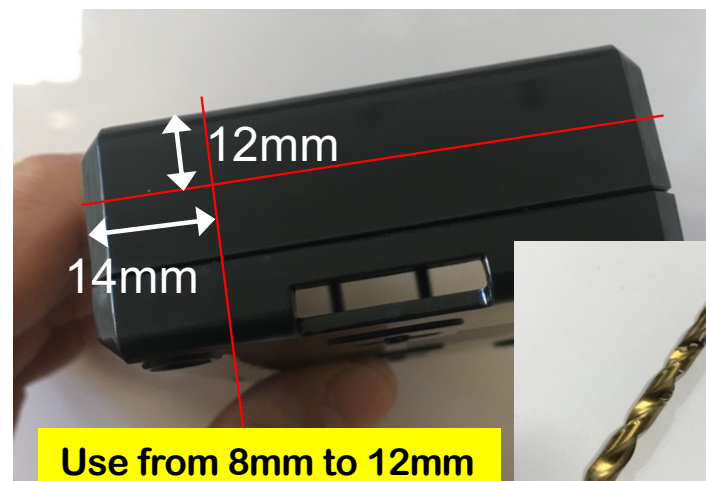
Using WAZIUP LoRa gateway hat



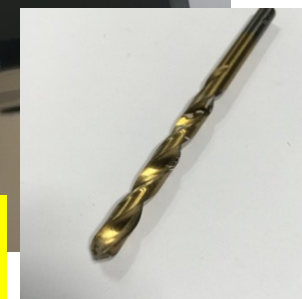
LoRa gateway hat
with on-board RTC,
no need for external
RTC module



An OLED screen can
directly be connected
to the radio hat



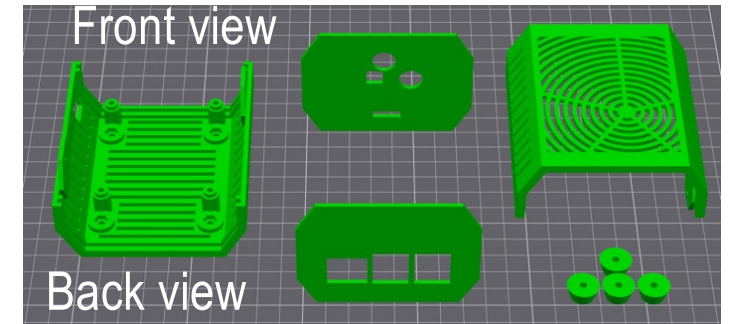
Use from 8mm to 12mm
drill bit for metal





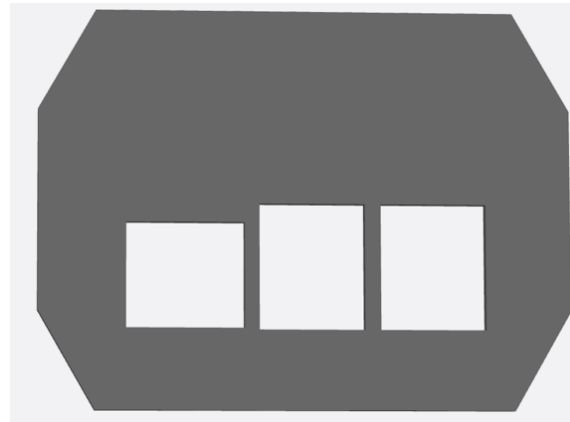
Using the Thingiverse case

<https://www.thingiverse.com/thing:3012012>

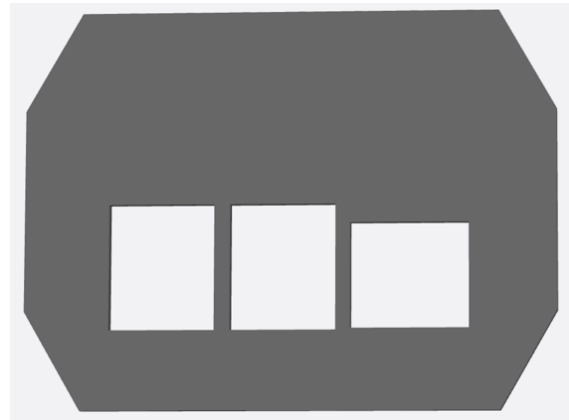
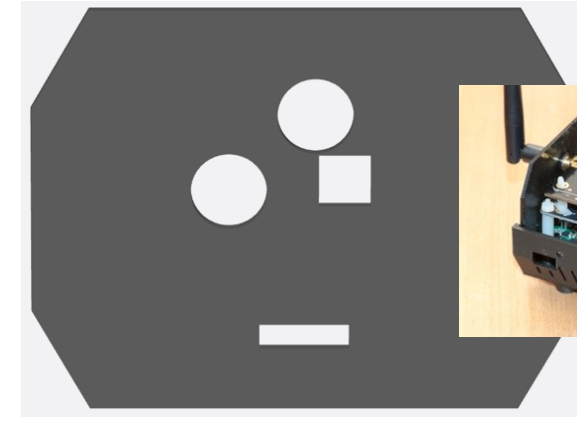


RPI3B/3B+

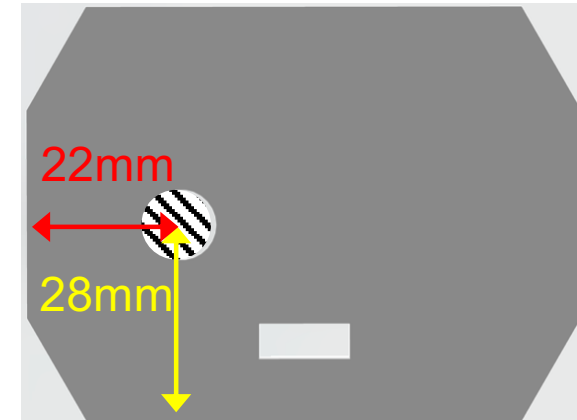
back panel
For RPI3B/3B+



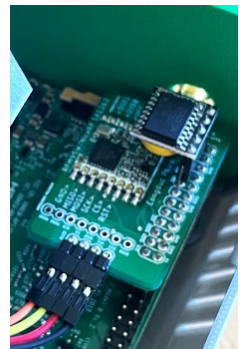
Front view: initial front panel designed for RAK831



Simply flip the back panel for RPI4B

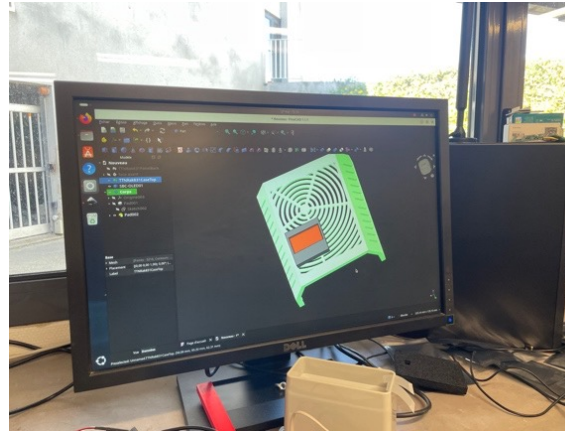
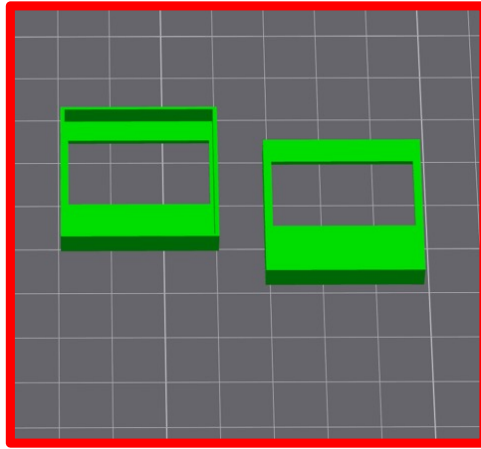


We provide a modified front panel





OLED cover with the Thingiverse case



6cm x 8cm
sticker





Multi-channel/Multi-SF

- It is possible to have a multi-channel/multi-SF gateway by connecting a LoRa radio concentrator such as the RAK2245 or the new RAK5146 with the RAK5146 Pi Hat (right photo)
- The LoRa radio concentrator will be automatically detected
- IMPORTANT:** a reliable 5V 2.5A power source is necessary



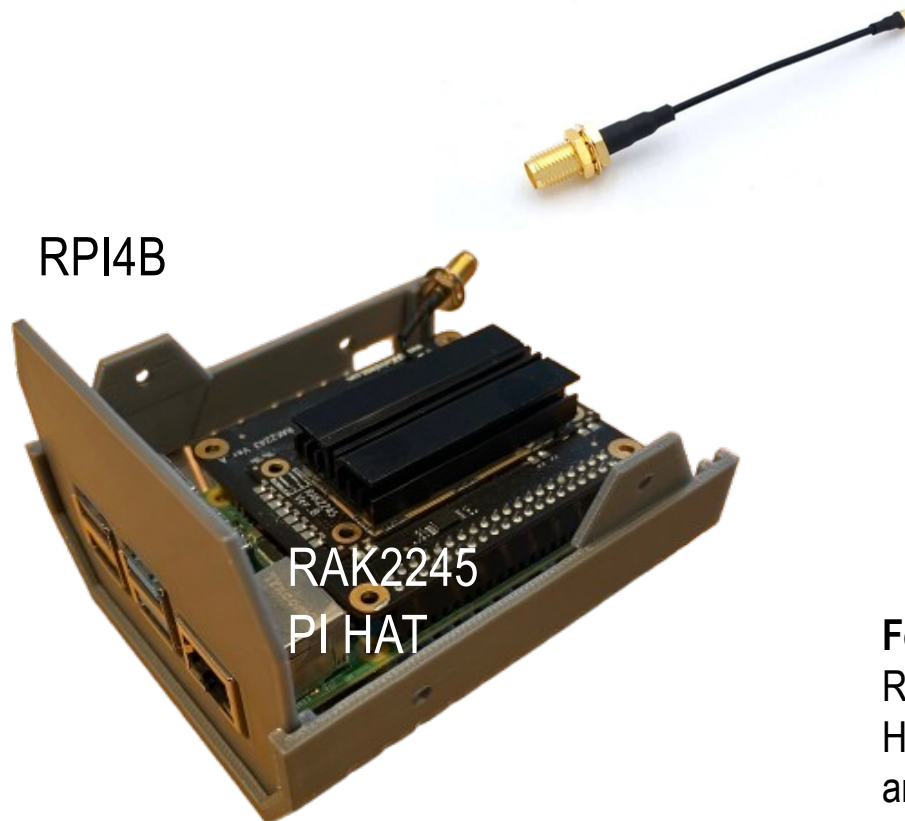
RAK2245



RAK5146



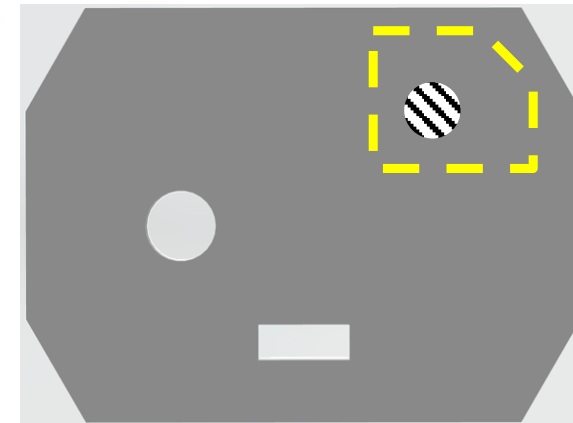
Thingiverse enclosure with multi-channel



RPi4B

RAK2245
PI HAT

Front view



For the multi-channel radio concentrator such as the RAK2245 or the new RAK5146 with the RAK5146 Pi Hat, simply drill a 8mm hole somewhere in the top-right area of the front panel for the ufl-SMA pigtail



Boot with mSATA SSD

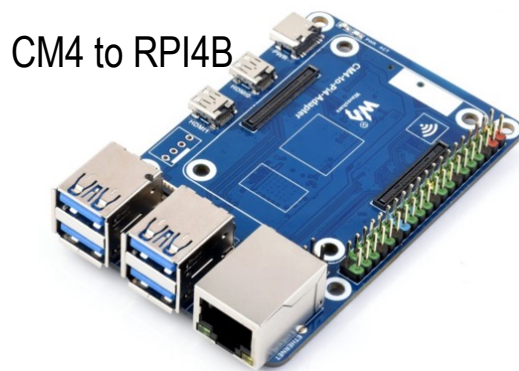
- For more reliability, the SD card image can be flashed on an mSATA SSD disk (with an mSATA-USB adapter) to avoid SD card issues
- 16GB is sufficient





Compute Module 4 with eMMC

- Or, use a Compute Module 4 with an adapter board to have an RPI 3B/3B+/4B based on eMMC to avoid SD card issues

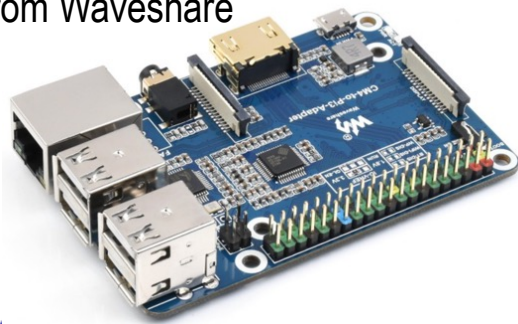


CM4 to RPI4B



CM4

RPI adapter boards
from Waveshare



CM4 to RPI3B/3B+



Mini Base A with aluminium casing

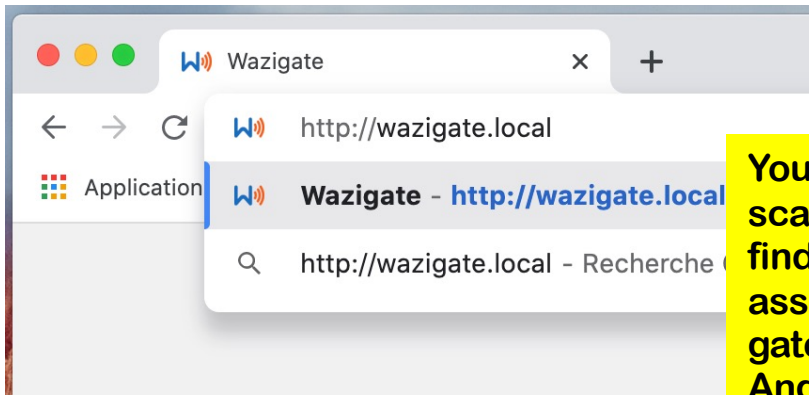






Checking the gateway (Ethernet cable)

- ⦿ Connect the gateway to your laptop which has Internet
- ⦿ Enable Internet sharing, laptop provides IP address to gateway
- ⦿ Power the gateway, wait 3-4mins for boot process
- ⦿ Open web navigator. Go to <http://wazigate.local> or use IP address



You can use an IP scanner program to find the IP address assigned to your gateway (such as Angry IP Scanner)

Wazigate Login to the Wazigate dashboard

Username*
admin

Password*
.....

LOGIN

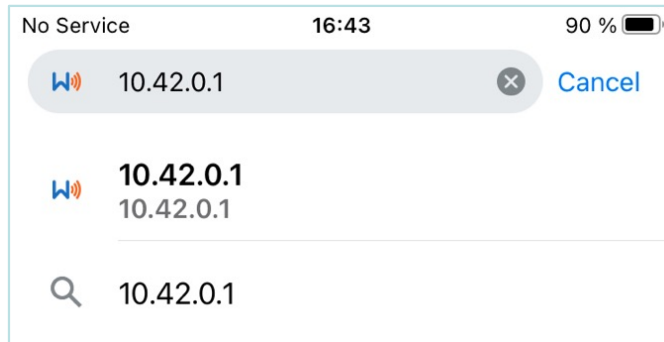
Default username: **admin**
Default password: **loragateway**

- ⦿ Use default login to connect
 - ⦿ User: admin
 - ⦿ Password: loragateway

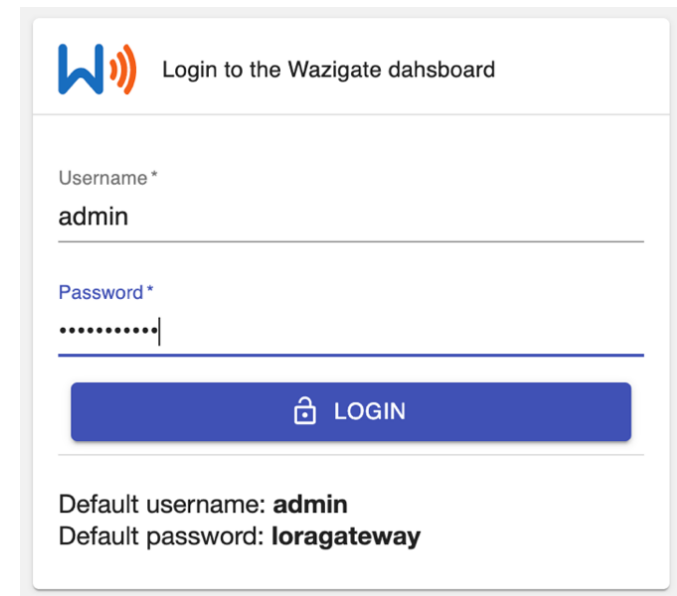


Checking the gateway (WiFi)

- ④ Use a smartphone to check access to gateway through WiFi
- ④ Connect to WAZIGATE_XXXXXXXXXXXXX WiFi network
 - ④ default WiFi password is loragateway
- ④ Open web navigator. Go to <http://wazigate.local> or <http://10.42.0.1>



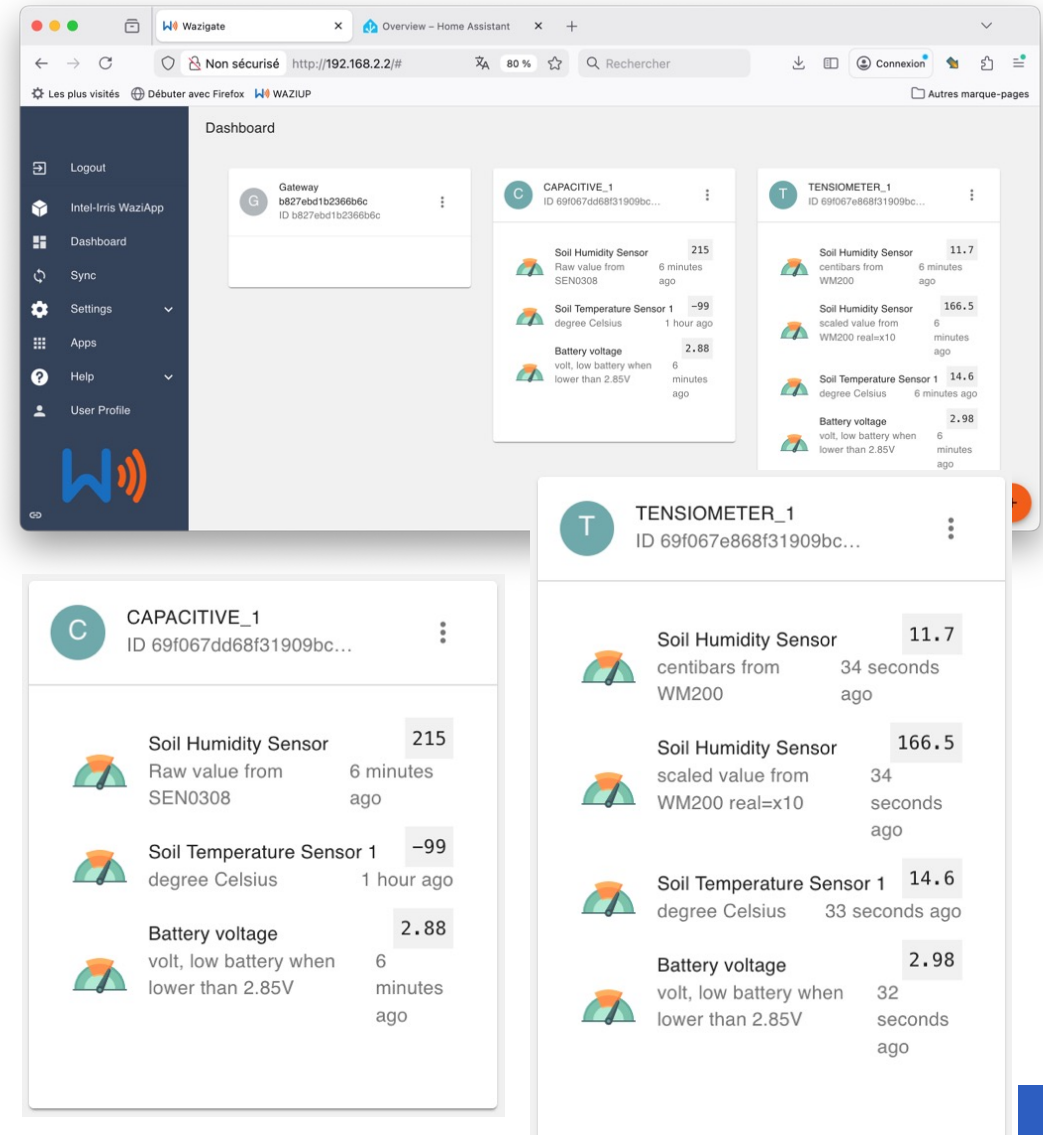
- ④ Use default login to connect
 - ④ User: admin
 - ④ Password: loragateway





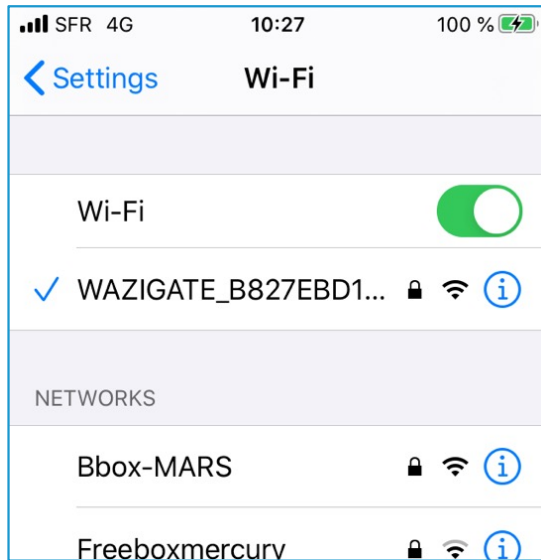
Default gateway configuration

- ④ The default configuration will be ready for
 - ④ 1 capacitive sensor named CAPACITIVE_1 with address 26011DAA
 - ④ 1 tensiometer sensor named TENSIMETER_1 with address 26011DB1
- ④ Capacitive device will show humidity, soil temperature (if any) and battery level values
- ④ Tensiometer device will show centibar, raw resistance, soil temperature and battery level values





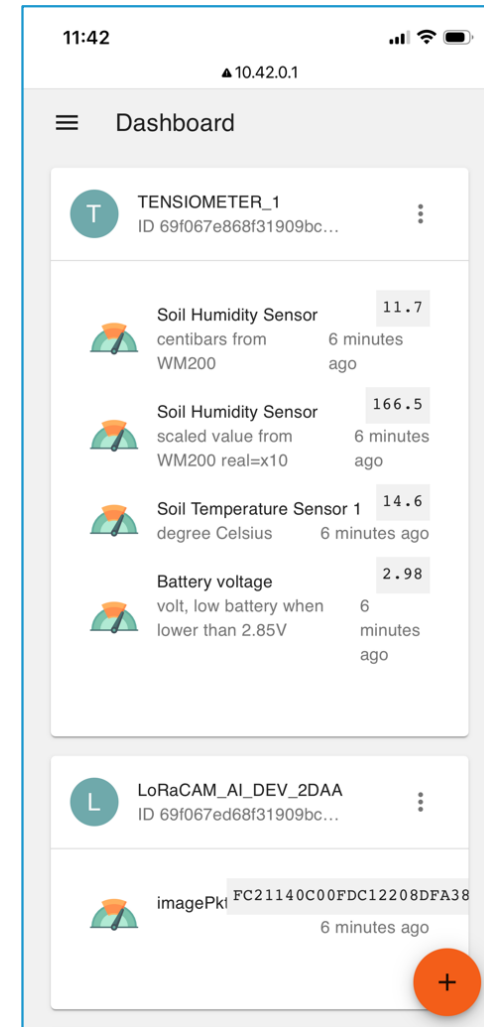
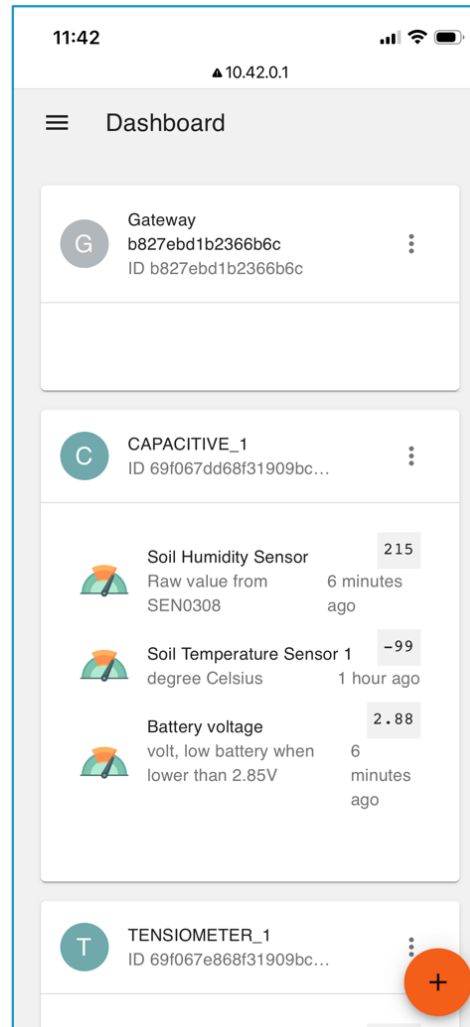
From a smartphone



<http://wazigate.local>



<http://10.42.0.1>





Sensor information on OLED

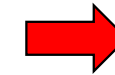
- ⦿ The OLED displays the latest received sensor data for the end-user: the device name, the time of last received data, the sensor raw value and the soil condition
- ⦿ The main screen is displayed for 6s every 30s. Then a screen saver display will show a shorter version of these information with a 5-bar visual
- ⦿ 5 bars: saturated | 4 bars: wet
- ⦿ 3 bars: wet | 2 bars: dry
- ⦿ 1 bar: dry | 0 bar: very dry





QR code for connecting to WiFi

- The gateway WiFi is WAZIGATE_XXXXXXXXXXXX where XXXXXXXXXXXXXXX will be the MAC address of the RPI
- For instance, WAZIGATE_B827EBD1B236
- A QR code for joining the WiFi network is displayed for 10s before the main screen to join with a smartphone
- Then, users can scan static QR code to connect to the gateway's dashboard
- OLED cycling pattern



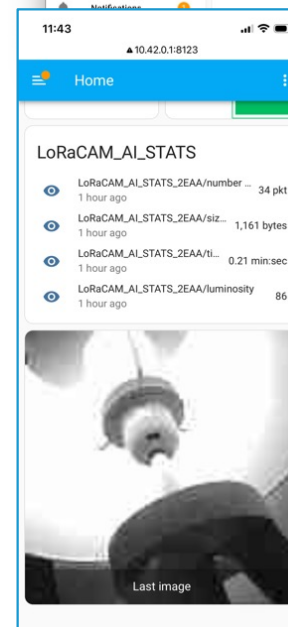
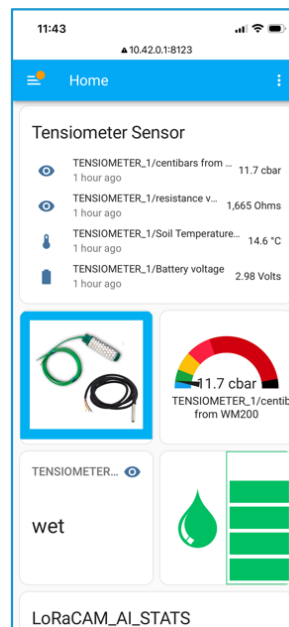
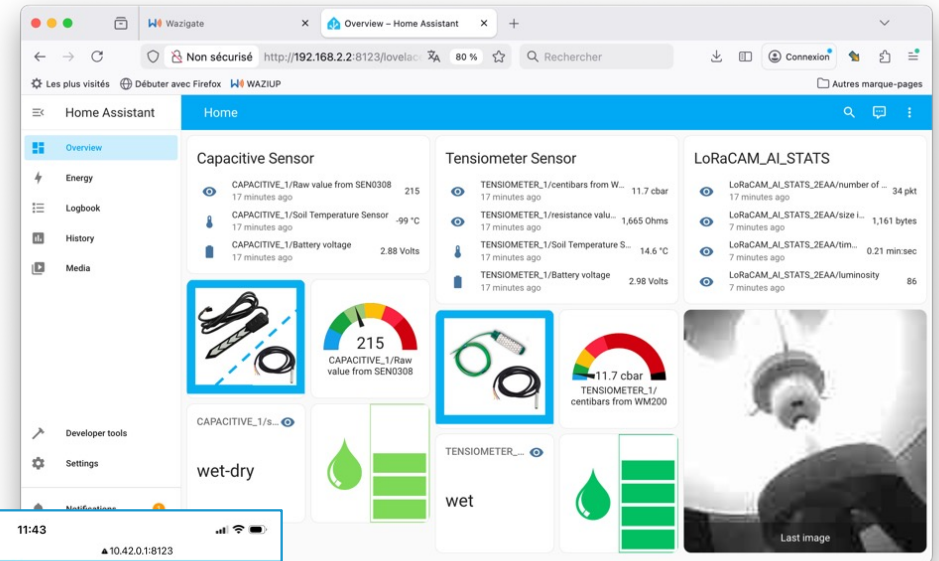
<http://wazigate.local> <http://10.42.0.1>





Home Assistant dashboard

- ⦿ Connect to
 - ⦿ <http://wazigate.local:8123> (Ethernet)
 - ⦿ <http://10.42.0.1:8123> (gateway's WiFi)
- ⦿ The HA login page should appear
 - ⦿ User: intelirris
 - ⦿ Password: intelirris





Connect to ChirpStack Network Server

- The generic WaziGate comes with ChirpStack Network Server
- Open web navigator. Go to <http://wazigate.local:8080> or use IP address

ChirpStack Login

Username / email *

admin

Password *

LOGIN

- Use default login to connect
 - User: admin
 - Password: admin
- Then get access to
 - registered devices
 - live LoRaWAN packet capture
 - downlink, ...

ChirpStack Application Server

Non sécurisé http://192.168.0.32:8080/#/organizations/1/applications/1

ChirpStack

Search organization, application, gateway or device

admin

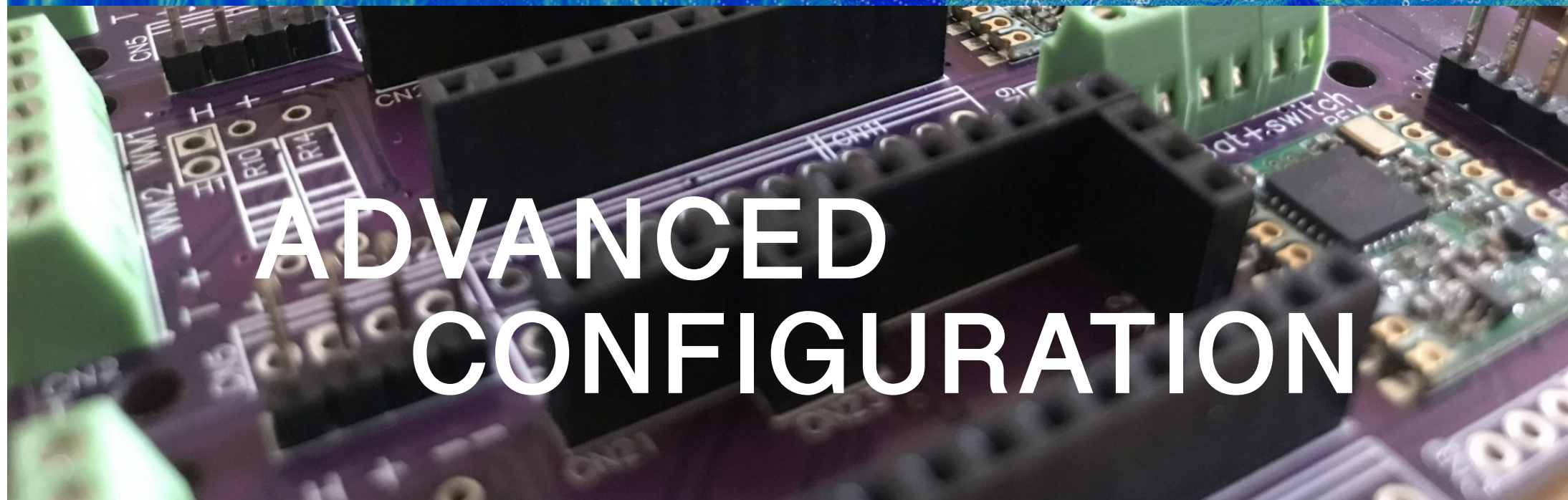
Applications / Wazigate

DEVICES APPLICATION CONFIGURATION INTEGRATIONS

+ CREATE

Last seen	Device name	Device EUI	Device profile	Link margin	Battery
a year ago	AA555A0026011DAA	aa555a0026011daa	Wazidev	n/a	n/a
n/a	AA555A0026011DB1	aa555a0026011db1	Wazidev	n/a	n/a
a year ago	AA555A0026012DAA	aa555a0026012daa	Wazidev	n/a	n/a
a year ago	AA555A0026012EAA	aa555a0026012eaa	Wazidev	n/a	n/a

Rows per page: 10 1-4 of 4

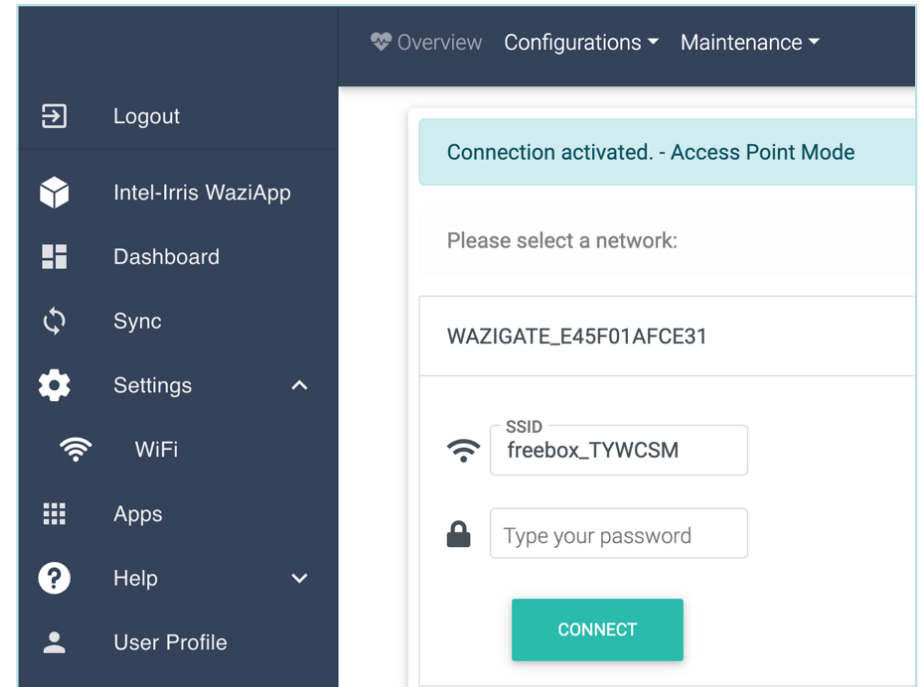




Advanced configuration

connect gateway to a WiFi network

- By default, the gateway acts as a WiFi Access Point
- To connect the gateway to a WiFi network, go to Setting/WiFi to list all available WiFi networks
- Then select the one you want in order to provide the WiFi password
- Ex: connect to freebox_TYWCSM
- Once connected, gateway is in WiFi Client mode





Advanced configuration

connect gateway to a WiFi network, con't

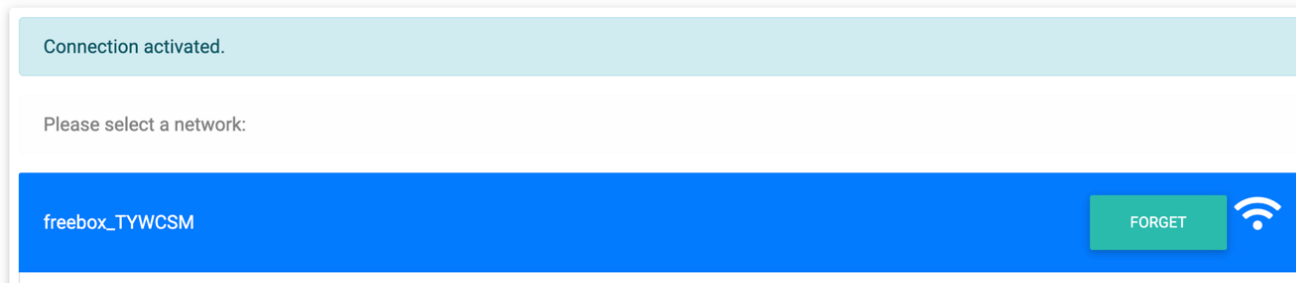
- ⦿ You can connect to several WiFi networks, one after another, to have a list of known WiFi networks
- ⦿ They will be memorized and if the current WiFi network is not available, another available network in the list of known WiFi networks will be selected
- ⦿ If there are no available WiFi networks in the list of known WiFi networks anymore, then the gateway switches back to Access Point mode



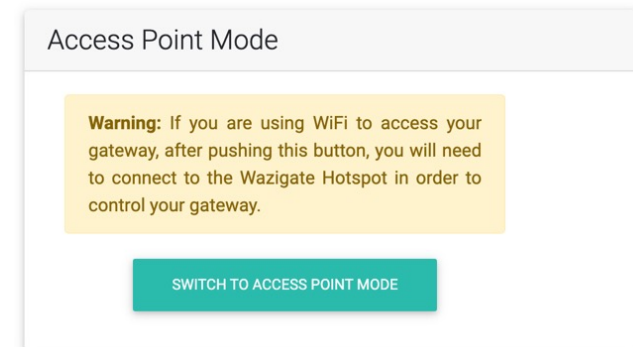
Advanced configuration

switch gateway back to WiFi access point mode

- ⦿ To get back to Access Point mode, got to Setting/WiFi and simply click on "Forget" for the current WiFi network



- ⦿ If you previously added several WiFi networks, click on "Forget" for **ALL** known & memorized WiFi networks
- ⦿ **DO NOT USE** the "SWITCH TO ACCESS POINT MODE" option
- ⦿ **IT IS NOT STABLE!**

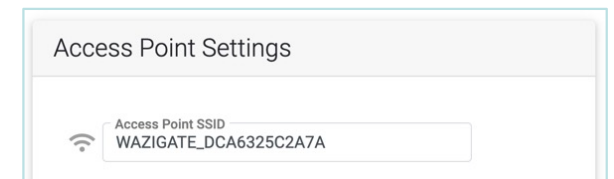
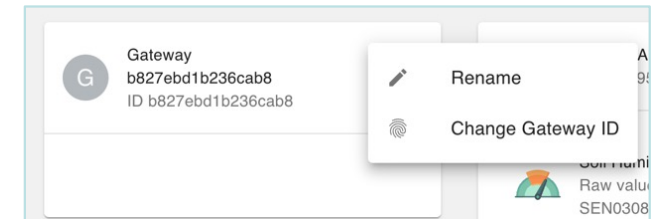




Advanced configuration

Synching your gateway to the cloud

- If you want to sync your gateway to the WAZIUP Cloud (WaziCloud), look at this tutorial page
 - <https://www.waziup.io/documentation/wazigate/v2/install/#registration-with-the-cloud>
- You will need an account on WaziCloud
 - If you don't have one, you need to create one first
 - <https://dashboard.waziup.io/>
- Then, you **NEED** to change your gateway id
 - Use the unique MAC address of your gateway that appears in Settings/Configuration (it is used as your gateway's WiFi hotspot)
 - Here: DCA6325C2A7A
 - Add 0000 at the end to have 16 digits
 - -> DCA6325C2A7A0000

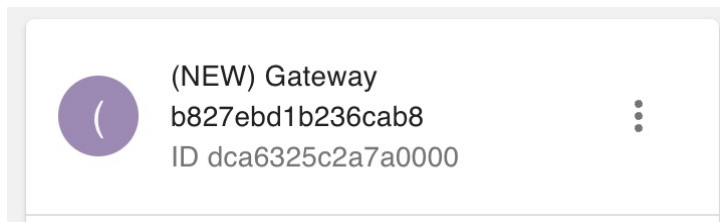




Advanced configuration

Synching your gateway to the cloud, con't

- ⦿ You should have a new gateway on your dashboard with the new ID



- ⦿ Enter your WaziCloud credentials in the Sync menu
- ⦿ Then, just activate sync on your gateway which needs to be connected to Internet
- ⦿ Log in the WaziCloud dashboard and check that you see your gateway and your dev
- ⦿ You can activate/deactivate synchronization at anytime



Advanced configuration

MQTT integration from WaziCloud

- ⦿ With sensor data on WaziCloud, it is possible to subscribe to those data with MQTT protocol
- ⦿ With command line `mosquitto_sub`
 - ⦿ `mosquitto_sub`
-L "mqtt://api.waziup.io/devices/<deviceId>/sensors/<sensorId>/value"
 - ⦿ `mosquitto_sub`
-h api.waziup.io -t devices/<deviceId>/sensors/<sensorId>/value
- ⦿ With other MQTT integration client/platform
 - ⦿ Host: api.waziup.io
 - ⦿ Topic: devices/<deviceId>/sensors/<sensorId>/value
- ⦿ Output
 - ⦿ `{"value":34,"timestamp":"2022-12-23T10:23:54Z"}`



Advanced configuration

MQTT integration – MQTT client on a smartphone

- Example with an MQTT client (EasyMQTT) on an iPhone7



Advanced configuration

MQTT integration – MQTT client on a smartphone

- Example with an MQTT client (EasyMQTT) on an iPhone7



Advanced configuration

MQTT integration – MQTT client on a smartphone

- Example with an MQTT client (EasyMQTT) on an iPhone7

Connect

MQTT BROKER

Name wazicloud

Host api.waziup.io

Port port number

Client ID EasyMQTT

Username username

Password password

MQTT 3.1.1 MQTT 5.0

TLS/SSL ☐

CONNECTED TO API.WAZIUP.IO

Disconnect ⚡

Connect Publish Subscribe EasyMQTT Plus Settings

Subscribe Messages

NUMBER OF MESSAGES

10 25 100

MESSAGES

2022-12-23 11:23:54

Topic: [devices/63a336da68f31908471dae68/sensors/temperatureSensor_0/value](#)

Message: {"value":34,"timestamp":"2022-12-23T10:23:54Z"}

Connect Publish Subscribe EasyMQTT Plus Settings